

Voice-driven UxS - Applicant Guide

Voice-Driven Command & Control for Uncrewed Systems

Overview

In modern operations, the ability to command uncrewed systems quickly and naturally is a force multiplier. What if an operator could simply *speak* to control drones — and the system would understand, validate, and execute?

Your task: build a voice-driven command and control (C2) system that translates natural speech into drone commands, enforces safety rules, and gives the operator clear feedback. You will work with simulated drones in a 3D environment, a pre-defined battlespace with waypoints and entities, and an automated scoring system that tests your system's accuracy and robustness.

Event Date: Saturday, March 21, 2026

Location: SAIT, Calgary

Event Time: 8:30 AM - 8:00 PM

Team Size: Up to 5

The Scenario

You are an operator at a forward operating base. You have a few drones on the pad and a battlespace with designated waypoints and known contacts — some friendly, some unknown, some hostile. Your sensor feeds and intelligence reports are available as structured data files.

Your job is to build a system where you speak commands — "Take off to 20 meters", "Fly to Waypoint Bravo", "Investigate the unknown contact" — and the drones respond. But it's not just about moving drones around. Your system must be *smart*: it should refuse to engage friendly forces, warn you about unknowns, confirm high-risk actions, and handle the messy reality of natural speech.

The core question: can your system turn voice into controlled, safe action?

What You're Building

Your system connects three layers: a voice pipeline that turns speech into text, a command engine that parses and validates those commands, and a drone interface that executes them in simulation.

Core System (Required)

1. Voice-to-Text Pipeline

Capture spoken commands and convert them to text using any speech-to-text engine. OpenAI Whisper (runs locally, no API key needed), Google Speech API, or the browser's built-in Web Speech API all work. The choice is yours.

2. Command Parser & Safety Engine

This is the intellectual core of the challenge. Parse natural language into structured drone commands. Validate them against safety rules: check the battlespace entity list before allowing engagement actions, require confirmation for high-risk commands, and block actions that target friendly forces.

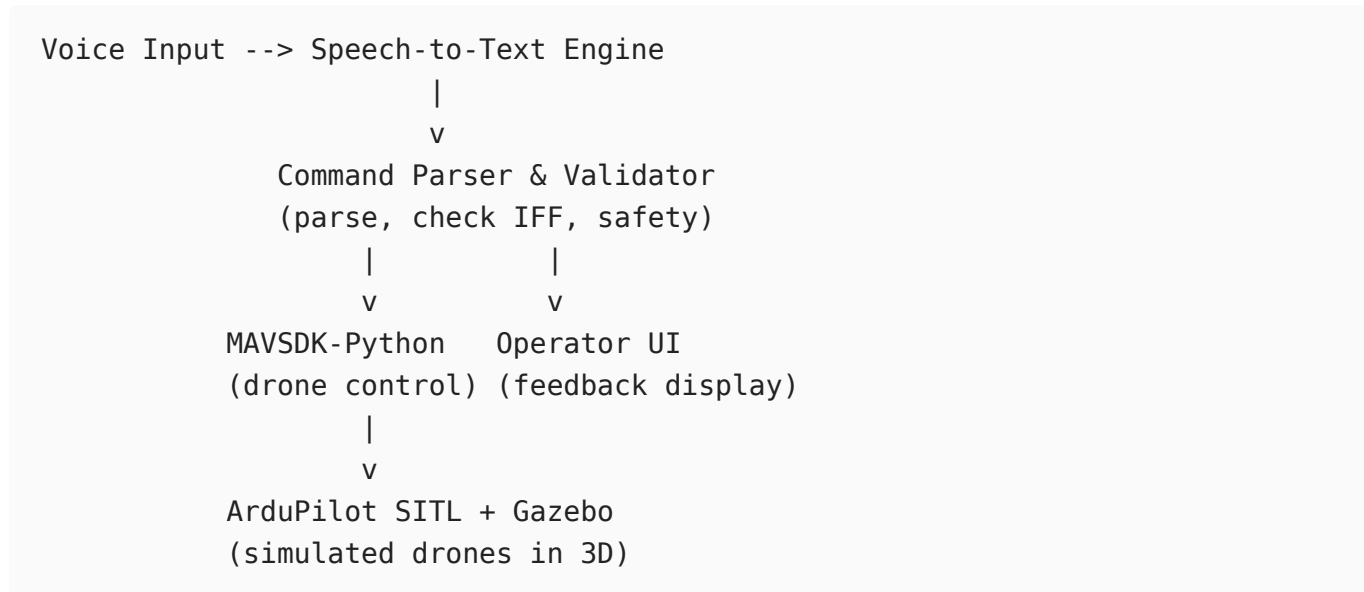
3. Drone Control Interface

Send validated commands to simulated drones via MAVSDK-Python or similar. Your drones run in ArduPilot SITL (Software-In-The-Loop) — a full flight controller simulation. You'll see them fly in a 3D Gazebo environment rendered in your browser.

4. Operator Feedback

Show the operator what's happening. At minimum: a command log (what was spoken, what was parsed, what was executed), drone telemetry (position, altitude, battery, mode), and clear confirmation prompts for high-risk actions. This can be a terminal, a web dashboard, or anything visual.

Suggested Architecture



The Battlespace

You will receive a set of data files describing the operational environment:

Waypoints — Named locations in the area of operations (Alpha through Hotel). Each has GPS coordinates. Commands like "fly to Waypoint Bravo" should resolve to the correct location.

Entity List (IFF) — A JSON file containing known contacts in the battlespace, each classified as FRIENDLY, UNKNOWN, or HOSTILE. Your system must check this list before executing engagement-related commands:

- Commands targeting **friendly** entities --> **Block** with a clear warning
- Commands targeting **unknown** entities --> **Require confirmation** before proceeding
- Commands targeting **hostile** entities --> Allow, but still confirm high-risk actions like "engage"

Drone Configuration — Two drones are available: "Alpha" and "Bravo". Each runs as a separate SITL instance on a known port.

These files will be available for download on the day of the event.

How You'll Be Evaluated

Evaluation details are being finalized and will be shared closer to the event.

Technical Score — Your system will be evaluated on how well it handles a range of commands, from basic single-drone control to more complex scenarios involving multiple drones, IFF safety logic, and edge cases. Expect your system to be tested on things like:

- Basic drone commands (takeoff, navigation, landing)
- Compound and parameterized commands
- IFF awareness and safety rule enforcement
- Multi-drone coordination
- Graceful handling of ambiguous, contradictory, or invalid input

Demos — Each team gets a short presentation slot to demo their system live and explain their design decisions. This is your chance to show off voice integration, UX polish, and creative features.

A practice dataset will be provided at the start of the hackathon so you can test your system during development. The final evaluation may use a different set of scenarios.

Technical Details

What You'll Receive on the Day

- **Cloud simulation environment** — A GPU-accelerated VM per team running Gazebo (3D visualization) + ArduPilot SITL (drone physics). Access via browser. Your code will control the SITL systems over the internet.
- **Starter code** — Connection examples, waypoint coordinates, battlespace entity file, and a practice dataset for self-testing during development
- **This guide** and the full challenge brief with evaluation criteria

Recommended Tools

Component	Recommended	Alternatives
Speech-to-Text	OpenAI Whisper (local)	Google Speech API, Web Speech API
Command Parsing	Python + regex or LLM	SpaCy, custom grammar
Drone API	MAVSDK-Python	pymavlink (lower-level)
UI Framework	Streamlit, Flask, terminal	React, PyQt, Tkinter
Map Display	Folium, Leaflet.js	Google Maps API

Technical Constraints

- **Software only.** No hardware is involved. Everything runs in simulation.
 - **Language agnostic.** Python is recommended but not required.
 - **Bring your own laptop.** You'll need a modern laptop with Python 3.10+ and a browser.
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Glossary

Term	Definition
UxS	Uncrewed Systems -- drones, rovers, and other unmanned platforms
C2	Command and Control -- the system and process for directing operations
SITL	Software-In-The-Loop -- a full flight controller simulation that runs on a PC
MAVSDK	The Python SDK for communicating with MAVLink-based drones and rovers
Gazebo	A 3D robotics simulator used to visualize drone flight
IFF	Identification Friend or Foe -- classifying contacts in the battlespace
RTL	Return to Launch -- a standard command for the drone to fly home and land
STT	Speech-to-Text -- converting spoken audio into written text
Waypoint	A named GPS coordinate in the area of operations
noVNC	A browser-based VNC client for accessing remote desktops

Timeline

Time	Activity
Pre-event	Applicant Guide Released (this document)
8:30 AM - 9:10 AM	Participant Check-In and Breakfast
9:15 PM - 9:45 PM	Kickoff & Challenge Reveal
9:50 PM - 12:15 PM	Hacking Session #1
12:15 PM - 1:00 PM	Lunch + Reset
1:00 PM - 4:00 PM	Hacking Session #2
4:00 PM - 4:30 PM	Final Submissions & Pitch Order Draw
4:30 PM - 4:45 PM	Break/Guest Arrival & Appetizers
4:45 PM - 5:00 PM	Opening Remarks and Introduction of Judges
5:00 PM - 6:45 PM	Final Presentations
7:10 PM - 8:00 PM	Awards & Close

Times are approximate and may shift. Final schedule will be confirmed closer to the event.
